UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Dipartimento di Ingegneria e Scienze
dell'Informazione e Matematica

Università degli Studi dell'Aquila

Introduction to

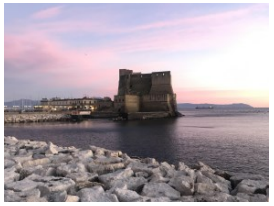
Artificial Intelligence - A.A. 2026/2027

[F4I] Master's Degree in Computer Science (AICoNDA)

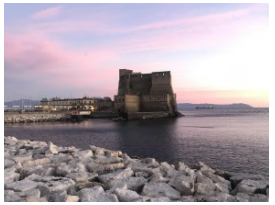
Prof. Fabio Persia, PhD

About myself… (1/3)

- Bachelor and Master degree
  - **Computer Engineering,** University of Naples "Federico II"



- PhD
  - **Computer Engineering,** University of Naples "Federico II", (Supervisors: Prof. Antonio Picariello and Prof. Vincenzo Moscato), 2013



  - **PhD Visiting Scholar** (under the supervision of Prof. V.S. Subrahmanian), University of Maryland Institute for Advanced Computer Studies (UMIACS), University of Maryland, College Park, MD, USA

About myself... (2/3)

- Department of Electrical Engineering and Information Technology, **University of Naples "Federico II"**
  - 2009/May -> 2009/Dec – Pre-doctoral Internship (Co.Co.Co.)
  - 2013/Jul -> 2015/Apr – Postdoctoral Research Fellow



- Faculty of Computer Science, **Free University of Bozen-Bolzano**
  - 2015/May -> 2018/Apr – Assistant Professor (RTD/A)
  - 2018/May -> 2020/Oct – Assistant Professor (RTD/A)



- DISIM, **University of L'Aquila**
  - 2020/Nov -> 2023/Oct – Tenure-Track Assistant Professor (RTD/B)



About myself... (3/3)



- Current Position
  - DISIM, **University of L'Aquila**

    2023/Nov -> ongoing –
    Associate Professor

- Research Interests
  - My research interests include the fields of **event detection** and **data analytics** in **large scale databases** applied to **multimedia**, **semantic analysis** and **security**

Learning Objectives

- The aim of the course is to provide students with an understanding of the **basic concepts** related to **artificial intelligence**, starting with an **introduction**, followed by a **historical overview**, and moving on to the **definition** and **use** of **intelligent agents**, **heuristic search**, and **predictive search**.

- The basic concepts related to **knowledge representation**, **probabilistic reasoning**, **probabilistic reasoning over time**, and **reinforcement learning** will also be studied.

Learning Outcomes

- Upon completion of this module, learners should have acquired concepts related to:
  - The characteristics of a system based on **artificial intelligence**
  - Modeling real-world problems using **intelligent agents**
  - The use of **knowledge representation**
  - Features and use of **heuristic search**
  - The use of **probabilistic reasoning**
  - The use of **reinforcement learning**

Prerequisites

- Basic knowledge of **data structures** and **programming** is required.

Contents (1/3)

- **Introduction:** What is AI, the Foundations of Artificial Intelligence, The History of Artificial Intelligence, The State of the Art, Risks and Benefits of AI.

- **Intelligent Agents**: Agents and Environments, Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents.

- **Knowledge Representation**: Ontological Engineering, Categories and Objects, Events.

Contents (2/3)

- **Quantifying Uncertainty:** Acting under Uncertainty, Basic Probability Notation, Inference Using Full Joint Distributions, Independence, Bayes' Rule and Its Use, Naive Bayes Models.

- **Probabilistic Reasoning:** Representing Knowledge in an Uncertain Domain, The Semantics of Bayesian Networks, Approximate Inference for Bayesian Networks.

- **Probabilistic Reasoning over Time**: Time and Uncertainty, Inference in Temporal Models, Hidden Markov Models.

Contents (3/3)

- **Reinforcement Learning:** Learning from Rewards, Passive Reinforcement Learning, Active Reinforcement Learning, Generalization in Reinforcement Learning, Policy Search, Apprenticeship and Inverse Reinforcement Learning, Applications of Reinforcement Learning.

Teaching methods

- Lessons and exercises with active student participation

Learning assessment

- Learning assessment includes:
  - a **written exam** (**50%**)
  - homework related to the part taught by Prof. Caianiello (**Homework_Caianiello, 25%**)
  - homework related to the remaining part of the course (**Homework_Persia, 25%**)
- The student's final grade is calculated as the **weighted average** of the scores obtained in the three parts mentioned above.

Written Exam

- Each question is assigned a **maximum score**.
- The student's final grade is calculated by adding up the scores obtained for each question.
- Specifically, students will be required to answer theoretical questions related to the **introduction** (10%), **intelligent agents** (15%), **knowledge representation** (15%), **quantifying uncertainty** (15%), **probabilistic reasoning** (15%), **probabilistic reasoning over time** (15%), **reinforcement learning** (15%)
- If the student's written assignment contains elements that are unclear, inconsistent, or otherwise prevent the instructor from making an objective assessment, the instructor may ask the student to discuss these aspects orally.

Homework_Persia

- This homework will be assigned during the course and must be submitted 3 days before the written exam.

- After the written exam, students will have the opportunity to discuss their homework orally.

Materials

- **Course slides** and (possible) **lesson-specific supplementary material** (e.g., lecture notes, papers, book chapters, tutorials, further slides, code examples, etc.).

- Slides will be as much **educational** as possible: they will have a format close to lecture notes.

Materials: Books

There is no official textbook for this course.

However, here is a (non-exhaustive) list of suggested books (course slides may be based on materials from some of them):

GENERAL BOOKS:

- Russell, Norvig, 2021 - **Artificial Intelligence: A Modern Approach**, 4th ed. (https://aima.cs.berkeley.edu/global-index.html)
- Aggarwal, 2021 **- Artificial Intelligence: A Textbook** (http://www.charuaggarwal.net)
- Frankish, Ramsey (2014) - The Cambridge Handbook of Artificial Intelligence (https://www.cambridge.org/core/books/cambridge-handbook-of-artificial-intelligence/3DCB2E04739722A99EDE86B7A34A30E3)

BOOKS ON AGENT-CENTRIC/SYMBOLIC AI:

- Poole, Mackworth, 2023 - **Artificial Intelligence: Foundations of Computational Agents**, 3rd ed. (https://artint.info/3e//html/ArtInt3e.html)

BOOKS ON MACHINE LEARNING:

- Aggarwal, 2023 - **Neural Networks and Deep Learning - A Textbook**, 2nd ed. (http://www.charuaggarwal.net)
- Mitchell, 1997 - **Machine Learning** (https://www.cs.cmu.edu/~tom/mlbook.html)

Office Hours

- Exploit lecturer's availability (office hours)
  - **Fabio Persia**
  - **Tuesdays**, 09:00 – 10:00 (please send me an email before)
  - **Office 215**, Blocco 0
  - Fabio.Persia@univaq.it
- Anytime, also on Microsoft Teams
  - Prior appointment

Course website

- Team named "**Artificial Intelligence A.A. 2026-27**"
- Team Code: **1xsl8i6**

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Good luck to everyone!

Eng. Fabio Persia, PhD

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